Exam for Data and Things, spring 2025

Oral exam info:

- There will 20 minutes for each student, including time for assessment and feedback
- At the beginning of the exam, the student draws a random number. Each number will correspond to an exam topic. (see the list of Exam topics below).
- The student then presents on the topic (3-5 min) including potential selected exercises handed in as part of the written product. This is followed by questions about the presentation and the exam question from the examiners (5-10 min).
- At the end, the examiners might relate their questions to some of the other exam topics.
- For each of the exam topics, the student is expected to know the central concepts, methods, theories, and problems discussed in class and be able to explain and exemplify them. Moreover, for those exam topics where there are selected exercises handed in as part of the written product, the student is expected to be able to explain how they solved the exercises and be able to explain their entire code.
- The oral exam is scheduled for the 20-21 of March.

Written product info:

- The written product will consist of answers to selected exercises – these will be selected from those that have already been done in class. Only some of the exam topics will have such selected exercises.
- The list of selected exercises will be made public on the last day of class (Monday March 10).
- The handed in answers to the selected exercises should be in the format of a single Jupyter Notebook or a zip file containing a notebook for each of the selected exercises.
- The students can do the selected exercises in self-made groups and either hand in as a group on Eksamen.ruc.dk, or hand in the notebook(s) individually.
- The hand-in must happen through eksamen.ruc.dk before March 16 at 10:00.

Exam topics:

1) Data transformation and exploratory data analysis (EDA)
2) Data engineering
3) Statistics
4) Regression
5) Time series
6) Classification
7) IoT and sensor data
8) Clustering
9) Machine Learning Operations (MLOps)
10) Neural networks and deep learning
11) Generative AI
12) Explainability
13) Ethical reflections on data science